Fluid Dynamics II

Notes made by Finley Cooper

12th October 2026

Contents

1 Introduction **3**

2 Equations for a Newtonian Viscious Flow **3**

2.1 Dynamic viscosity 3

2.2 Rate of strain tensor 3

2.3 The stress tensor 3

1 Introduction

The Part IB Fluid Dynamics course mainly covers inviscid flow and irrotational flow including potential flow. The course covered an introduction to viscious flow for the very simple case where $\mathbf{u} = (u(y,t), 0)$ in 2D parallel flow.

This course is about the effects of viscosity and answers the following very important questions

(i) How does viscosity affect the equations for a general flow?

(ii) What happens if viscosity dominates over the inertia of the fluid?

(iii) What happens if the flow is nearly paralllel?

(iv) What happens if viscosity is small compared to inertia?

(v) What about stability?

2 Equations for a Newtonian Viscious Flow

2.1 Dynamic viscosity

Recall from Part IB Fluid Dynamics we have Couette flow where a plate at rest and a plate with velocity U are parallel with a space h between them filled with a viscous fluid. For a simple fluid we find that the force required to move the top plate F is proportional to $\frac{uA}{h}$. Hence the shear stress is

$$\tau_s = \frac{F}{A} = \mu \frac{U}{h}$$

where μ is the dynamic viscosity. The shear stress is due to the relative motion between each layer of fluid and its neighbours above and below.

2.2 Rate of strain tensor

Consider the velocity near a fixed point $\mathbf{0}$ *wlog*. Taylor expand $\mathbf{u}$ so we have the multivaraible Taylor expansion

$$u_i(\mathbf{x}) = u_i(0) + x_j \left.\frac{\partial u_i}{\partial x_j}\right|_{\mathbf{0}} + \frac{1}{2} x_j x_k \left.\frac{\partial^2 u_i}{\partial x_j \partial x_k}\right|_{\mathbf{0}} + \cdots$$

We drop the quadratic terms and decompose $\frac{\partial u_i}{\partial x_j}$ into symmetric and antisymmetric parts so

$$\begin{aligned}\frac{\partial u_i}{\partial x_j} &= \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \\ &= e_{ij} + \frac{1}{2}\varepsilon_{jik}\boldsymbol{\omega}_k\end{aligned}$$

where e_{ij} is the *rate of strain tensor* and $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity. Hence we have that

$$\mathbf{u}(\mathbf{x}) \approx \mathbf{u}(\mathbf{0}) + \mathbf{e} \cdot \mathbf{x} + \frac{1}{2}\boldsymbol{\omega} \times \mathbf{x}.$$

We have that $e_{ij} = e_{ji}$ by definition so $\mathbf{e}$ is real symmetic and hence has real orthogonal eigenvectors which are the principle axis of strain. We have that $e_{ii} = \nabla \cdot \mathbf{u} = 0$ so the eigenvalues sum to 0. The eigenvalues are called the principle rates of strain.

2.3 The stress tensor